

Multi Agent Reinforcement Learning for Air Traffic Control

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Project report - October 2026

Abstract

We study reinforcement learning (RL) for single-aircraft navigation in the BlueSky-Gym air-traffic-control environment, where an Airbus A320 must reach a goal waypoint while avoiding polygonal restricted areas and staying inside a sector. Working on the environment, reward and evaluation side of a three-person team, we make five contributions. (i) We show that the environment’s stock reward ranks *circling safely forever* above *reaching the goal*, and fix the ranking by rebalancing the terminal reward. (ii) We design potential-based shaping functions [19] that are direction-aware and, crucially, *geodesic*: the potential is the length of the shortest obstacle-avoiding route, computed on a visibility graph, which removes the local-minimum traps that straight-line potentials create behind concave obstacles while preserving policy invariance. (iii) We build an evaluation methodology that exposes failure modes the default metric hides: a *clean-success* metric (goal reached with zero violations) reveals that a tabular agent with 99.3 % goal completion is clean in only 66.8 ± 2.3 % of held-out scenarios; multi-seed analysis with increasing sample sizes shows that apparent seed-to-seed differences were largely sampling noise; and we reproduce the official competition protocol exactly. (iv) We map more than twenty exploration methods to the observed failure modes and run a controlled study of exploration for PPO [27], finding that temporally correlated action noise (pink, Ornstein–Uhlenbeck) *hurts* at default scale, the entropy coefficient matters little, and warm-starting from demonstrations of a visibility-graph planner reaches 90 % goal completion after only 0.2 M environment steps; on the official 1000-episode protocol this policy is the cleanest agent we have (76.9 % clean vs. 67.4 % for the tabular agent, and twice its clean rate on hard maps). (v) Recorded trajectories on held-out maps show that the two agent families fail in opposite ways: the tabular agent always arrives but cuts through obstacles on more than half of the hard maps and reverses its turn direction on 41 % of decisions, whereas PPO policies are an order of magnitude smoother and rarely enter obstacles but time out when a long detour is required. We also report an evaluation pitfall: observation normalisation fitted without traffic silently corrupts a policy’s inputs when traffic appears, collapsing goal completion from 77 % to 7.5 %.

1 Introduction

Air traffic control (ATC) is a natural testbed for safe RL: the dynamics are smooth and well understood, but the agent must trade progress against hard safety constraints, and failures are easy to define and count. The AI4REALNET BlueSky-Gym competition [11], built on the open-source BlueSky simulator [14], asks for an RL policy that issues heading and speed commands to guide aircraft to a goal waypoint while avoiding other traffic, restricted areas and the sector boundary. Entries are scored on nine objective metrics over the first 1000 scenarios of a fixed random seed.

Our team split the work into three tracks: *environment and reward* (the author), *state representation* and *learning algorithms* (two teammates; see the acknowledgements). This report covers the author’s track and the evaluation and exploration studies that grew out of it. The starting point was a tabular agent (Q-learning, SARSA and Expected SARSA over a discretised egocentric state) that reached the goal almost always but routinely clipped restricted areas, oscillated between left and right turns, and—when a large or concave obstacle blocked the direct line—turned back,

circled, and eventually flew through the obstacle. Our project advisor’s review pointed to three directions: continuous states, policy optimisation, and better exploration. The last is the subject of the second half of this report.

Contributions.

1. **Objective diagnosis.** The stock terminal reward of +1 against per-second penalties makes never approaching the goal the highest-return policy; we quantify this and fix it (§3.1).
2. **Geodesic potential-based shaping.** A direction-aware potential and a visibility-graph geodesic potential that has no local minima behind concave obstacles, with an implementation that adds 0.3 ms per decision and is continuous by construction (§3.2).
3. **Evaluation methodology.** A clean-success metric, held-out scenario blocks, a multi-seed reliability analysis with an explicit outlier test, an exact reproduction of the official protocol, and metrics for turn oscillation and for hard scenarios (§4).
4. **Exploration study.** A taxonomy of exploration methods mapped to the observed failure modes, nine behaviour policies for the tabular agent, and a controlled PPO study of entropy regularisation, coloured noise and planner demonstrations (§5).
5. **Behavioural analysis.** Trajectories of four agents on the same held-out maps, selected by fixed rules, which separate violation-type failures (tabular) from commitment-type failures (PPO) and quantify turn oscillation (§6).

2 Background

Environment. Each episode samples a sector polygon, five polygonal obstacles (500–1500 NM² each), a start and a goal. The A320 flies at a fixed altitude; the agent acts every 10 s of simulated time. An episode terminates when the aircraft is within 5 km of the goal and is truncated after 3000 s. The official evaluation adds ten scripted intruders; a loss of separation is any approach closer than 5 NM. The nine scored metrics are goal completion, flight time, and event counts and durations for intrusions, restricted-area entries and sector exits. The reward is free for competitors to design and is not scored.

Tabular agents. Q-learning, SARSA and Expected SARSA [29] share a table and differ only in the bootstrap target ($\max_a Q(s', a)$, $Q(s', a')$ and $\mathbb{E}_{a \sim \pi} Q(s', a)$ respectively). The state is a discretisation of six egocentric features—goal drift and distance, bearing and clearance of the nearest obstacle edge, and bearing and distance to the sector boundary—into $8 \cdot 4 \cdot 8 \cdot 5 \cdot 8 \cdot 4 = 40,960$ states. Actions are heading changes of $\{-15, -7.5, 0, +7.5, +15\}^\circ$. The step size follows a Robbins–Monro schedule $\alpha = n(s, a)^{-0.6}$ [26].

PPO agent. Our teammate’s continuous-control track uses PPO [27] from Stable-Baselines3 [23] with a 186-dimensional observation (the shipped observation plus 55 hand-crafted features including 16 free-space rays), a two-dimensional action (heading change up to $\pm 15^\circ$ and speed change), 8 parallel simulators, rollouts of 8×1024 steps, two 256-unit hidden layers, and observation and reward normalisation.

Potential-based shaping. Adding $F(s, s') = \gamma \Phi(s') - \Phi(s)$ to the reward leaves the optimal policy unchanged for any real-valued Φ [19]. Two implementation rules matter in practice: Φ must be zero at true terminal states (otherwise a $\gamma^T \Phi(s_T)$ term rewards finishing at particular times), and truncation is not termination. A good Φ approximates the optimal value function; a poor one does not change the optimum but can slow or mislead learning.

3 Reward design

3.1 The stock objective prefers never arriving

The base environment pays +1 on capture and -1 per second for each kind of violation, i.e. up to -10 per decision. We computed discounted returns ($\gamma = 0.99$) of hand-written reference policies on fixed scenarios. A policy that circles safely inside the sector earns about +87, while a policy that actually flies to the goal earns about -60 : the highest-return behaviour is never to approach the goal. Disabling shaping leaves the ranking unchanged, so the defect is in the task reward, which shaping was faithfully preserving. Because a direct flight takes 67–113 decisions and $\gamma^{100} \approx 0.37$, the terminal reward must be large to survive discounting; the ranking flips for a terminal reward between 250 and 500, and we use 1000 for margin. In plain terms: reaching the goal is worth 1000 seconds of safety violation.

3.2 Direction-aware and geodesic potentials

Direction-aware potential. Our first potential combines progress, heading alignment and proximity, all in kilometres of equivalent detour and divided by the 1.5 km flown per decision:

$$-\Phi(s) \cdot 1.5 = d + w_\delta \frac{|\delta|}{180^\circ} d + (1 + \kappa(\beta_o)) c(c_o; 20, 8) + (1 + \kappa(\beta_b)) c(c_b; 20, 10), \quad (1)$$

where d is the distance to the goal, δ the drift from the goal bearing, c_o and c_b the signed clearances to the nearest obstacle and to the boundary, β_o, β_b their bearings relative to the heading, $\kappa(\beta) = (1 + \cos \beta)/2$ is 1 for a hazard dead ahead and 0 behind, and $c(x; w, s) = w e^{-x/s}$ for $x \geq 0$ and $w(1 + |x|/s)$ inside a violation, which keeps a gradient pointing back out. The bearing term makes turning away from a close hazard improve Φ , which a distance-only proximity cost cannot. Weights are $w_\delta = 0.3$; scales of 8–10 km are several decisions wide, avoiding cliffs that inflate temporal-difference targets.

The local-minimum problem. Every term in (1) starts from the straight-line distance d . Behind a large or concave obstacle, every step of the correct detour *increases* d , so shaping penalises the very manoeuvre that is needed—the classical failure of potential-field navigation [16]. Figure 1(a) shows greedy ascent of the straight-line potential driving straight into the pocket of a U-shaped obstacle; this matches the behaviour we observed in simulation (turn back, circle, fly through).

Geodesic potential. We replace d and δ by the length of the shortest obstacle-avoiding route and the heading error to that route’s first leg. Once per episode we inflate every obstacle by a margin $m = 2$ km, take the corners of the inflated shapes inside the sector as nodes of a visibility graph [18], connect mutually visible nodes and run Dijkstra from the goal, giving the remaining distance $D(v)$ from each node. At every decision, the route cost from position p with heading h is

$$\min_{v \in V(p) \cup \{g\}} (\|p - v\| + D(v)) \left(1 + w_\delta \frac{|\delta_v(h)|}{180^\circ}\right), \quad (2)$$

where $V(p)$ is the set of nodes visible from p and $\delta_v(h)$ the heading error towards v . Three details make the potential well behaved:

- *Visibility is tested against obstacles inflated by $0.99m$, not the originals.* With the originals, a far corner becomes visible before the line to it is tangent to the inflated shape and the route length jumps; a unit test found jumps of 1.2 km in a 0.5 km step before this fix.
- *The drift weight is applied before the minimum,* so when the preferred route switches sides (around the left versus the right end of a wall) the potential remains continuous.
- *Inside an obstacle* the route goes via the nearest exit. This is continuous on entry, so entering is never rewarded; it jumps only at an obstacle’s medial axis, inside a violation, where leaving by the far side has genuinely become the shorter violation.

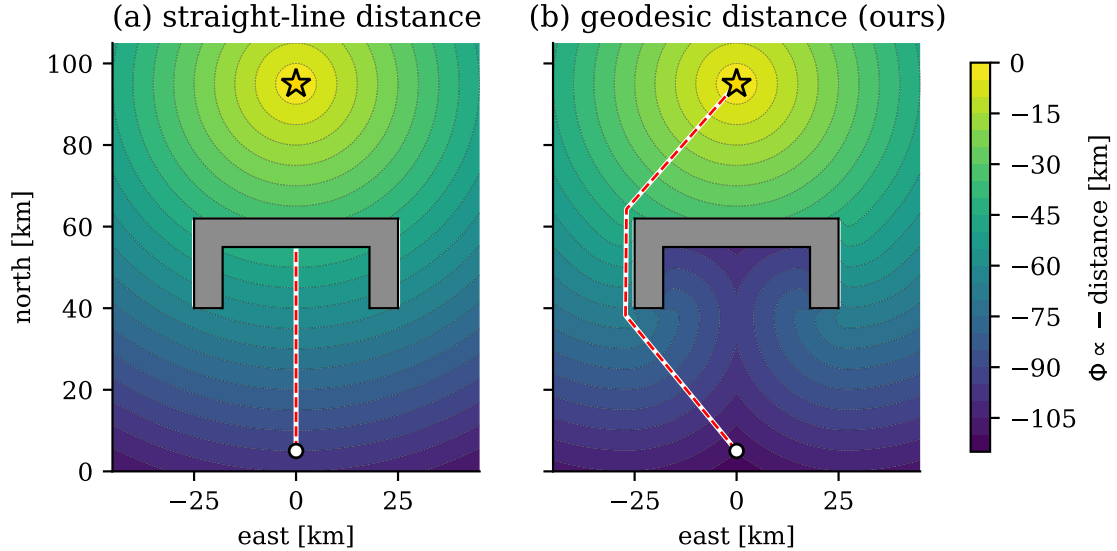


Figure 1: Potential around a concave obstacle (grey) with the goal (star) behind it. The dashed path is greedy ascent of Φ from the start (circle). (a) Straight-line distance has a local maximum inside the pocket. (b) The geodesic potential has none; ascent follows the shortest feasible route.

Because obstacles are fixed within an episode, the geodesic potential is a function of state and policy invariance holds. Only Φ 's input changes; the agent's state is untouched. In live simulation the overhead is 0.3 ms per decision (26.0 vs. 26.3 ms), and the route is more than 5 % longer than the straight line in 30.5 % of decisions (up to $2.7\times$), i.e. the correction is far from cosmetic. Eleven offline tests check, among other things, that the route is 1-Lipschitz along paths, that a concave pocket is not a local minimum, that a wall touching the sector edge forces the route around the other end, and that entering an obstacle never shortens the route. Training with this potential is in progress.

4 Evaluation methodology

Clean success. Goal completion alone counts an episode that clips an obstacle for 80 s and then arrives as a success. We define *clean success* as reaching the goal with zero restricted-area entries and zero sector exits.

Seeds and held-out scenarios. We distinguish the *training seed*, which fixes an agent's whole training stream, from *scenario seeds*, which fix individual test maps. Agents are evaluated on blocks of scenario seeds disjoint from any used for training or checkpoint selection.

Reliability across training seeds. We trained five Expected-SARSA agents with identical hyperparameters (3000 episodes, ε decayed linearly from 1 to 0.05 over 2400 episodes) and different training seeds, and evaluated each on held-out scenarios. Table 1 and Figure 2 show the results. Goal completion is 99–100 % for every seed, but clean success is only 62–76 %. Increasing the held-out sample from 100 to 300 scenarios shrank the across-seed standard deviation of clean success from 4.7 to 2.3 points: much of the apparent seed-to-seed spread was sampling noise, not genuine differences between agents. A boxplot-style outlier test (below $Q_1 - 1.5\text{IQR}$), applied only with at least four seeds because the minimum of any sample is always *someone*, flags no failing seed (fence 59.3 %, minimum 63.7 %). Evaluating one agent on 1000 held-out scenarios gives 98.8 % goal completion, 67.5 % clean success and 70.1 s in restricted areas per episode—larger than the 30.7 s that the first 100 scenarios suggested for the same agent. Small held-out samples were optimistic, which is why all reported numbers below use at least 300 scenarios unless stated otherwise.

Table 1: Expected SARSA, five training seeds, held-out scenarios (no intruders). Clean = goal reached with zero restricted-area entries and zero sector exits.

Seed	100 held-out maps			300 held-out maps		
	Goal [%]	Clean [%]	Restr. [s]	Goal [%]	Clean [%]	Restr. [s]
0	100.0	65.0	45.0	100.0	63.7	66.9
1	99.0	68.0	32.7	99.0	69.3	55.8
2	100.0	68.0	42.8	99.7	66.0	62.8
3	99.0	62.0	56.5	99.0	65.3	65.8
4	100.0	76.0	30.7	99.0	69.7	60.2
Mean $\pm$ sd	99.6	67.8 $\pm$ 4.7	41.5	99.3 $\pm$ 0.4	66.8 $\pm$ 2.3	62.3

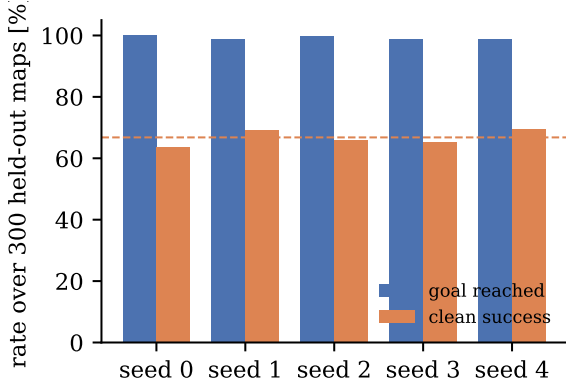


Figure 2: Goal completion hides a one-in-three violation rate. Dashed: mean clean success.

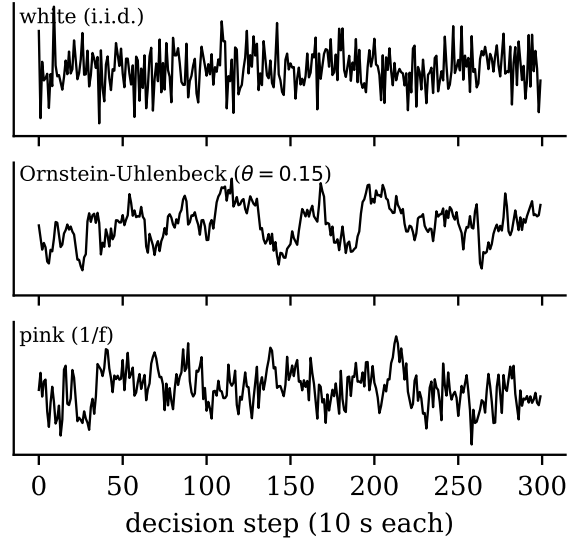


Figure 3: Exploration noise sequences with unit variance. Correlated noise holds a turn for many decisions; white noise cancels out.

Official protocol. The competition scores the first 1000 episodes of seed 42, seeding only the first reset so the random stream then continues, with ten intruders. We reproduce this exactly for both agent families and verified that our environment subclass generates the same scenario stream as the official environment (identical goal positions and obstacle counts in each of the first episodes compared). We add two measurements: *reversals per 100 decisions* (consecutive turn commands of opposite sign, the oscillation our advisor highlighted) and clean success on *hard scenarios*, those whose obstacle-avoiding route at spawn is more than 1.1 times the straight-line distance.

An evaluation pitfall: normalisation without traffic. The PPO agents were trained without intruders, so the intruder block of their observation was constantly zero and the running normaliser recorded near-zero variance for those inputs. At evaluation with ten intruders, real values are divided by that near-zero variance and clipped at ± 10 : the network receives out-of-distribution inputs. For the demonstration policy, goal completion on the official scenarios was 7.5 % over 1000 episodes without the fix and 76.7 % on the first 30 with it. Since the tabular agent has no intruder features, we evaluate traffic-free PPO policies *intruder-blind*—intruders are present and scored, but the policy sees the zero block it was trained on—which puts both families on the same footing. The intrusion metrics of all agents in this report therefore measure flying blind through traffic and are a floor, not a result.

Table 2: Exploration methods considered. Tab = usable with the tabular learner; PG = usable with PPO. F1: sustained detour; F2: distant gap.

Family	Methods	Tab	PG	Target
Undirected, i.i.d.	ε -greedy, Boltzmann, Gaussian action noise	✓	✓	—
Temporally correlated	εz -greedy [5], OU [17], pink noise [6, 15]	✓	✓	F1
Consistent per episode	parameter noise [22], NoisyNets [10], gSDE [24]	✓ ^a	✓	F1
Optimism / uncertainty	optimistic init., UCB [3], MBIE-EB [28], bootstrapped ensembles [20]	✓	ens.	F1, F2
Intrinsic motivation	RND [4], ICM [21], DIAYN [8], Go-Explore [7]	counts	✓	F2
Entropy	entropy bonus, SAC [12]	soft- Q	✓	—
Structure	curricula [9], demonstrations [13, 25], HER [2], shielding [1]	✓	✓	F1, F2

^a as a per-episode perturbation of the Q -table.

5 Exploration

5.1 Which failures can exploration address?

The two dominant failures are (F1) a large obstacle across the direct line, which requires tens of seconds of consistent turning *away* from the goal, and (F2) a distant narrow gap, which requires committing to a far target over a nearer, blocked one. Under per-step ε -greedy exploration, k consecutive same-direction random turns occur with probability about $(\varepsilon/5)^k$, so the training data almost never contain a successful detour. Exploration can only change *what is learned*; evaluation is greedy. It cannot compensate for information missing from the state (the tabular state sees only the nearest point of the nearest obstacle, so a distant gap is invisible), nor for a shaping term that penalises the detour—hence the geodesic potential of §3.2.

Table 2 summarises the methods we considered and how they map to F1/F2 and to our two learner families.

5.2 Tabular behaviour policies

We implemented nine behaviour policies behind a single interface, all paired with Q -learning: because its target is $\max_a Q(s', a)$ it learns the greedy value correctly under any behaviour policy, whereas SARSA and Expected SARSA assume the behaviour is ε -greedy—Expected SARSA’s closed-form target is simply wrong for correlated or demonstration-driven behaviour. The arms are ε -greedy; εz -greedy (random actions held for a zeta-distributed duration, capped at 12 decisions); a per-state gradient bandit whose preferences are updated from the TD advantage; Ornstein–Uhlenbeck and pink noise added to the greedy heading change and rounded to the nearest action; per-episode Gaussian perturbation of the Q -table with an adaptive scale that tracks the ε schedule (the tabular analogue of parameter noise and NoisyNets); optimistic initialisation; UCB on visit counts; and demonstrations, in which a decaying share of training episodes is flown by a planner following the geodesic route. A tenth arm repeats ε -greedy with the straight-line potential to isolate the effect of the geodesic shaping. All arms pass smoke tests; the full ten-arm, three-seed study is queued (§8).

One finding came out of the design itself: the original code comments called the zero-initialised Q -table optimistic, which was true when returns were negative. After the terminal reward was raised to 1000, learned values are positive, so untried actions now look *worse* than tried ones and initialisation actively discourages exploration.

5.3 PPO exploration study

Arms. With all other settings fixed (one training seed, 1 M steps, default PPO hyperparameters) we compared: the Gaussian policy with the default entropy coefficient 0.005; entropy coefficients

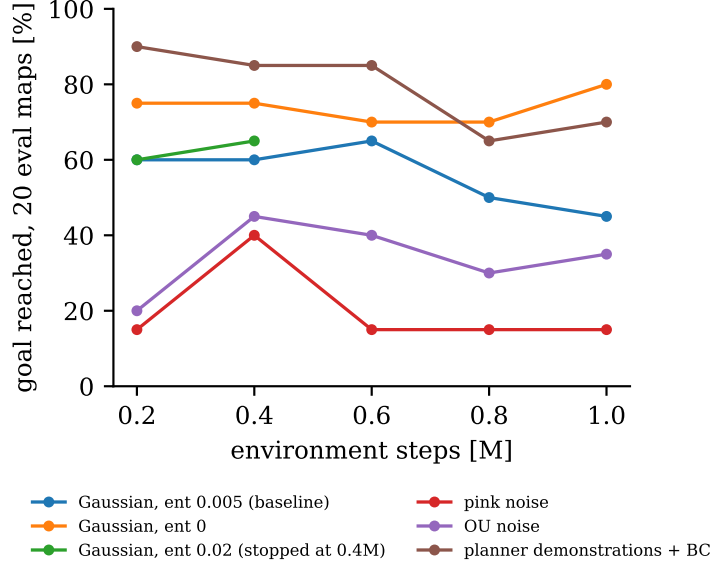


Figure 4: PPO exploration arms: goal completion on 20 fixed evaluation maps during training (one seed each; standard error ≈ 10 points).

0 and 0.02; pink and Ornstein–Uhlenbeck ($\theta = 0.15$) noise; gSDE with the exploration matrix resampled every four decisions; RND with a bonus annealed to zero over 0.5 M steps (novelty also rewards exploring obstacle interiors); and a demonstration warm start in which 100 episodes flown by the visibility-graph planner are cloned into the policy mean before PPO begins. Coloured noise is injected by replacing the i.i.d. standard-normal draw in $a = \mu(s) + \sigma\epsilon$ with a per-environment sequence that is regenerated at every episode boundary. Each step’s marginal stays $\mathcal{N}(0, 1)$, so the Gaussian log-likelihoods used in PPO’s ratio remain correct (Figure 3). Per-episode weight perturbation is not used with PPO because it makes the likelihood ratio incorrect; gSDE is the on-policy member of that family.

Results during training. Figure 4 shows goal completion on 20 fixed evaluation maps every 0.2 M steps (standard error ≈ 10 points, one seed per arm, so only large differences are meaningful). Three effects are large enough to report:

- **Correlated noise hurt.** Pink and OU noise reached at most 40–45 % and ended at 15 % and 35 %; episodes increasingly timed out. With the default initial $\sigma \approx 0.6$, a correlated perturbation holds a $\sim 9^\circ$ turn bias for many consecutive decisions, producing long wandering arcs. The effective exploration scale of coloured noise is much larger than that of white noise at the same σ , so these arms are likely under-tuned rather than refuted.
- **Entropy: an apparent effect that did not survive.** Removing the entropy bonus gave the best final training-time result (80 % vs. 50 % for the default), but on the official protocol the gap shrinks to three points (Table 3), well within single-seed noise. The training-time gap was an artefact of 20 evaluation maps that were also used for checkpoint selection.
- **Demonstrations gave a large head start.** The warm-started policy reached 90 % goal completion at the first evaluation (0.2 M steps), the best value of any arm at any time, then drifted to 70 % as PPO updates moved it away from the cloned behaviour—the regime in which keeping a decaying demonstration loss (DAPG [25]) is expected to help.

On the first 30 official scenarios, evaluated intruder-blind, the demonstration policy reached the goal in 76.7 % of episodes, all of them clean, with zero restricted-area time and only 2.2 turn reversals per 100 decisions.

Table 3: Official protocol: seed 42, first 1000 episodes, ten intruders (policies trained without traffic; PPO evaluated intruder-blind). Hard = route at spawn $> 1.1\times$ straight line (28 % of episodes). Rev. = turn reversals per 100 decisions. Intr. = loss-of-separation events per episode, flying blind. Expected SARSA: mean $\pm$ sd over five training seeds; PPO: one seed per arm.

Agent	Goal [%]	Clean [%]	Hard clean [%]	Restr. [s]	Flight [s]	Rev.	Intr.
Expected SARSA	99.4 ± 0.2	67.4 ± 1.1	25.2 ± 1.4	71.8 ± 4.8	1056	40.3	1.3
PPO, Gaussian ent 0.005	56.4	55.4	28.5	10.5	2167	3.8	1.8
PPO, Gaussian ent 0	59.3	57.8	31.0	22.3	2086	6.9	1.7
PPO, pink noise	18.2	17.8	4.7	19.3	2753	1.2	1.8
PPO, OU noise	25.0	23.0	8.7	30.9	2671	0.6	1.8
PPO, planner demos + BC	79.7	76.9	50.5	20.5	1772	2.1	1.7

Table 4: Four agents on the same 150 held-out maps (no intruders; PPO uses each run’s best checkpoint). Hard = route at spawn $> 1.1\times$ straight line (47 maps). Rev. = turn reversals per 100 decisions.

Agent	Goal [%]	Clean [%]	Hard clean [%]	Restr. [s]	Flight [s]	Rev.	Time-outs [%]
Expected SARSA (tabular)	100.0	74.0	44.7	36.9	1057	40.6	0.0
PPO + planner demos	78.7	76.7	57.4	1.7	1770	2.2	21.3
PPO, entropy 0	61.3	61.3	36.2	13.5	2077	7.1	38.7
PPO, pink noise	18.7	18.7	4.3	25.2	2700	1.1	81.3

Official protocol. Table 3 reports all agents on the official 1000-episode protocol. The ranking seen during training largely holds, with one exception. The demonstration-initialised PPO policy is the cleanest agent: 76.9 % clean against 67.4 % for the tabular agent, twice its clean rate on hard maps (50.5 % vs. 25.2 %), and less than a third of its time inside obstacles (20.5 s vs. 71.8 s). It pays with roughly 20 % time-outs and longer flights (1772 s vs. 1056 s), the commitment failure analysed in §6. Coloured noise remains clearly harmful (18–25 % goal completion). The exception is entropy: the 30-point training-time gap between entropy 0 and the default collapses to 3 points here. The five tabular seeds agree closely (standard deviation 1.1 points on clean success), confirming that the tabular result is a property of the method rather than of a lucky seed. All agents incur 1.3–1.8 losses of separation per episode because none can see traffic; the tabular agent’s lower count reflects shorter flights, not avoidance.

6 Behaviour in simulation

To see *how* the agents fail, we recorded every decision of four agents on the same 150 held-out maps (scenario indices 1000–1149, never used for training or checkpoint selection, no intruders) and drew the actual BlueSky trajectories. Of these maps, 47 (31 %) are *hard*: the obstacle-avoiding route at spawn is more than 1.1 times the straight line. Table 4 summarises the episodes; Figures 5 and 6 show individual ones, chosen by fixed rules stated in the captions rather than by hand.

The two families fail in opposite ways. The tabular agent *always* arrives (100 %) but pays for it in safety: it is clean on only 44.7 % of hard maps and spends 36.9 s per episode inside obstacles; in the episodes shown it does so by forcing a passage through gaps between obstacles that are too narrow (Figure 5c, 6). The PPO policies are the mirror image: they rarely enter obstacles, but when the direct line is blocked they fail to commit to the detour and run out of time (21–81 % time-outs, Figure 6). This is precisely failure F1 of §5, seen from the other side: what the tabular agent lacks is a penalty strong enough to make cutting through unattractive, and what PPO lacks is experience of completing long detours. The demonstration-initialised policy, whose warm start contains such detours, is the only agent that beats the tabular agent on clean success for hard maps (57.4 % vs. 44.7 %).

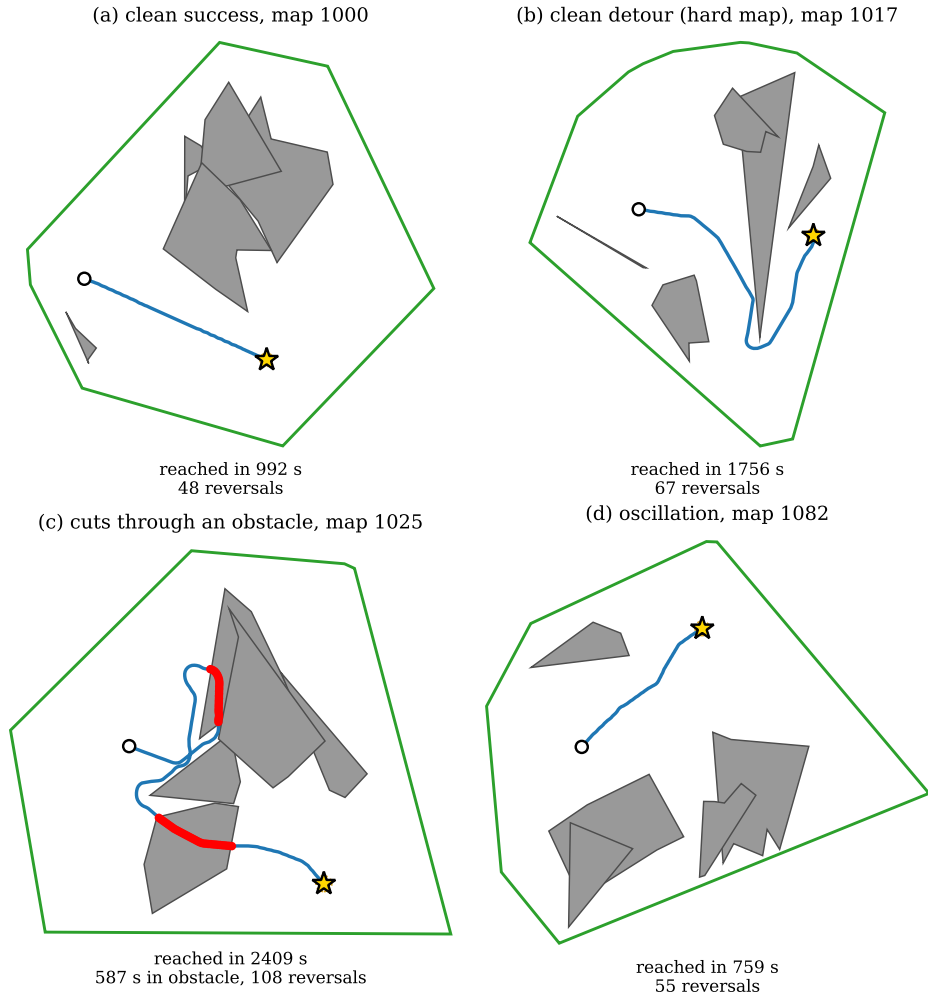


Figure 5: Recorded trajectories of the tabular Expected-SARSA agent (start: circle, goal: star with its 5 km capture radius, obstacles: grey, sector: green, red dots: decisions spent inside an obstacle). Panels are selected by rule from the 150 held-out maps: (a) the clean success with the most direct route, (b) the clean success with the longest detour, (c) the episode with the most time inside obstacles, (d) the episode with the most reversals per decision. Even the straight flight in (a) reverses its turn command 48 times: alternating $\pm 7.5^\circ$ commands cancel in the track but not in the commands.

Oscillation. The tabular agent reverses its turn direction on 40.6 of every 100 decisions; the PPO policies on 1–7. A likely structural cause is that with five discrete heading changes the only way to hold a heading that falls between them is to alternate, e.g. $\pm 7.5^\circ$, whereas a continuous action can simply output a small value. If so, this supports the advisor’s suggestion to move to continuous actions independently of any exploration effect.

Selection bias in training-time evaluation. The entropy-0 policy reached the goal on 80 % of the 20 maps used during training to choose its best checkpoint (Figure 4), but on 61.3 % of the 150 held-out maps. Checkpoint selection on a small fixed set overstates performance, which is why the comparisons in this section and in Table 3 use only scenarios that played no part in training or selection.

7 Engineering notes that affected results

Three infrastructure issues changed what could be measured and are worth recording for reproducibility.

- **Thread oversubscription.** With default settings the PPO learner process occupied about 20 CPU cores spin-waiting in PyTorch’s thread pool on a 256×256 network, starving the simulator workers. Restricting PyTorch to one thread raised throughput to about 110 environment steps per second with eight simulators.
- **Failures under heavy concurrency.** Starting 27 simulator processes at the same moment crashed the workers, and later runs lost workers when more than about 40 processes shared a 32 GB machine; memory exhaustion is the most likely cause. Staggered starts fixed the first problem; the second ended the gSDE and RND runs early and is the reason those arms are missing from the results.
- **Output buffering.** A long run that appeared hung was in fact healthy: Python block-buffers standard output when redirected to a file. Unbuffered output (`python -u`) is now used for all runs.

8 Limitations and future work

- **Single seed for PPO.** Each PPO arm was trained once; only the large effects above should be read as findings. The planned confirmation is three seeds for the four most informative arms (entropy 0, default, demonstrations, pink noise at reduced σ).
- **Incomplete arms.** gSDE and RND runs failed for infrastructure reasons and the entropy-0.02 run stopped at 0.4 M steps; the tabular exploration study and training with the geodesic potential are implemented and tested but not yet run at scale.
- **No traffic during training.** All policies are blind to intruders. The PPO observation already has a fixed-width intruder block so obstacle-only policies can be fine-tuned with traffic; doing so, and then moving to the multi-agent environment, is the main remaining step for the competition.
- **State limits of the tabular agent.** The tabular state contains only the nearest obstacle, without its size, so distant gaps are invisible regardless of exploration—one of the motivations for the continuous-state track.
- **Next experiments.** Geodesic potential for the PPO reward; DAPG-style decaying demonstration loss; coloured noise with σ scaled to match white noise’s displacement; a scenario-difficulty-dependent terminal reward that compensates for the extra discounting of long detours.

Acknowledgements

This work was carried out in a three-person team. Aditya implemented the tabular agents and identified the Robbins–Monro step-size schedule as the change that made seed-to-seed comparisons

meaningful; Sidharth implemented the geometric state features and the PPO training pipeline used in §5. The author was responsible for the environment and reward (§3.1–3.2), the evaluation methodology (§4), and the exploration study (§5). We thank our project advisor for the review that shaped the second half of this work.

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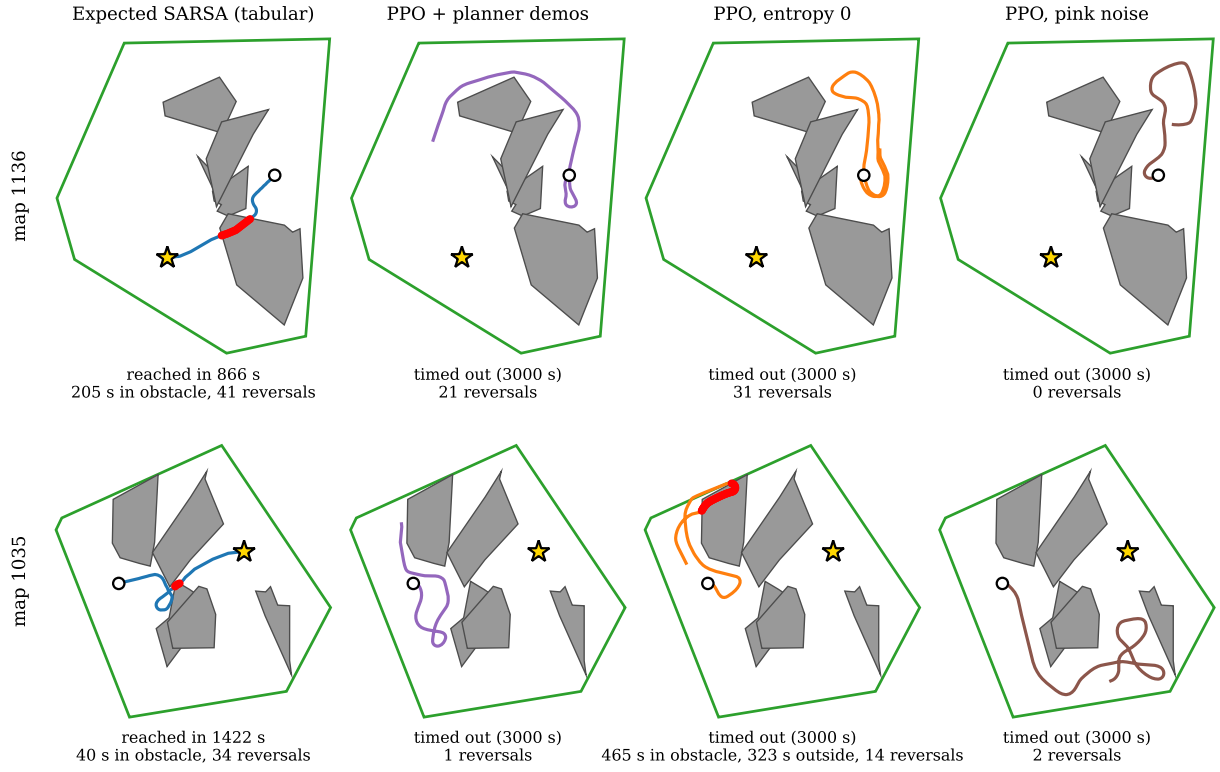


Figure 6: The same two hard maps flown by four agents. The maps are the hard maps (route $> 1.1 \times$ straight line) on which the tabular agent committed a violation, ranked by route ratio. The tabular agent reaches the goal by cutting through the gaps between obstacles; every PPO policy turns away from the obstacles but never commits to a detour, loops, and times out.